

UEL-CN-7000 Mental Wealth; Professional Life (Dissertation)

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Dissertation

Title: Accuracy vs. Efficiency: A comparative evaluation of Pre-trained CNN architectures for pneumonia detection from Chest X-rays

Abstract

Pneumonia remains a leading cause of mortality among children under the age of five globally, especially in underdeveloped countries where healthcare facilities are under-resourced. Access to radiologist and diagnostics infrastructure is severely limited. Deep learning particularly convolutional neural networks (CNNs), has shown promise for automated pneumonia detection from chest X-ray images. Previous studies have managed to prove that, CNNs can accurately detect pneumonia, with a higher accuracy than human radiologist. However they focused predominantly on classification accuracy neglecting computational efficiency, a critical aspect for consideration when deploying models in low resourced settings.

This study addresses this gap by presenting a comparative evaluation of three prominent pre-trained CNN architectures, VGG16, ResNet50 and EfficientNet for pneumonia detection from chest X-rays. A transfer learning was used on public available Kaggle chest X-ray dataset (5216), with a 70/15/15 stratified split for training, validation and testing. To ensure fair comparison, all three models were trained under identical hyperparameters, evaluation conducted on (1) classification accuracy and (2) computational efficiency. Evaluation metrics such as accuracy, precision, recall, F1-score and confusion matrix were used to compare predictions of the three models, whereas CPU inference time and peak memory usage was used to gauge computational efficiency. ResNet50 attained the highest test accuracy of 98.09%, followed by VGG16 at 97.96% and EfficientNet at 92.47%. In terms of computational efficiency, EfficientNet was the most resource efficient model (29 MB, ~15 ms, ~392 peak memory usage) , VGG16 was the slowest (528 MB, ~47 ms ,~1100 MB) peak memory usage.

This study contributes a practical, data driven guide for choosing a CNN architecture, to use in low resourced health care settings. ResNet50 is recommended for standard workstation, used in hospitals and clinical environment, while EfficientNet is better suited for edge devices and mobile deployment where storage and speed are crucial. Finally the study also demonstrate that deep learning models when deployed appropriately, provide reliable and efficient medical diagnostic support in resource constrained environments, ultimately contributing improving patient outcomes.

Declaration of Authenticity

I, Shepard Kudzaiishe Munyaradzi, declare that the content of this dissertation is the result of my own original work. To the best of my knowledge, this work has not been submitted, in whole or in part, for the award of any degree or for any other purpose.

I confirm that all sources and support used in its preparation have been fully acknowledged and cited.

Shepard Kudzaiishe Munyaradzi

Date: 07/09/2026

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
AP	Anterior-Posterior
AUC-ROC	Area under the Receiver Operating Characteristic Curve
BCE	Binary Cross-Entropy
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CT	Computed Tomography
CUDA	Compute Unified Device Architecture
CVPR	Conference on Computer Vision and Pattern Recognition
FLOPs	Floating Point Operations
FN	False Negative
FP	False Positive
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
HIPAA	Health Insurance Portability and Accountability Act
ICML	International Conference on Machine Learning
IDE	Integrated Development Environment
ICLR	International Conference on Learning Representations
JPEG	Joint Photographic Experts Group
MAC	Memory Access Cost
MB	Megabyte
ML	Machine Learning
MLP	Multilayer Perceptron

ms	Millisecond
NIH	National Institutes of Health
NPU	Neural Processing Unit
RAM	Random Access Memory
ReLU	Rectified Linear Unit
SGD	Stochastic Gradient Descent
TN	True Negative
TP	True Positive
VRAM	Video Random Access Memory
WHO	World Health Organization

1. Introduction

1.1 Research Area and Significance

Pneumonia is an acute respiratory disease of the lung whereby the air sacs become inflamed, leading to accumulation of fluids and even pus in the air sacs. Pneumonia, as per the World Health Organization, is the most common killer in children across the world accounting for 14 percent of all the deaths in children aged below five years (World Health Organization, 2022). The incidence rate of pneumonia is higher in low and middle income nations due to lack of adequate medical imaging services and radiologists.

The chest x-ray continues to be the most commonly used modality for diagnosing pneumonia. Yet, accurate assessment of the chest x-ray demands a high degree of professional skill and years of practical experience. The shortage of radiologists in resource-poor environments results in a bottleneck for accurate diagnosis.

1.2 Background of the Research Problem

With deep learning and specifically convolutional neural networks (CNN), there has been great progress in analyzing medical images. In one experiment conducted in 2017, a CNN was able to achieve the same level of accuracy in diagnosing pneumonia using chest X-rays compared to a radiologist(Rajpurkar et al., 2017) . This finding made it possible to think about automated systems for prioritizing the most urgent cases and making diagnoses in cases where a radiologist is not available.

In transfer learning, a CNN pre-trained using a big dataset like ImageNet is fine-tuned for the intended application, especially when there is little data available to train new models. Transfer learning is a feasible technique to use since it gives good results despite the small amount of data used (Mabrouk et al., 2024).The three popular architecture techniques include VGG16, famous for its simplicity and depth(Simonyan & Zisserman, 2015); ResNet50, which has skip connection, thus allowing extremely deep networks(He et al., 2016); and EfficientNet, which has compound scaling(Tan & Le, 2020).

1.3 Statement of the Research Problem

Although there have been some researches comparing VGG16 and ResNet50 in detecting pneumonia (Dharmik, 2025) and some others comparing ResNet50 and EfficientNet (Filvantorkaman & Torkaman, 2025), there is not yet any research comparing all three networks VGG16, ResNet50, and EfficientNet on the same footing (Asnaoui et al., 2020). From the systematic literature review it is apparent that there exists a lack of information, which makes it difficult to make the appropriate decisions in favor of a certain pre-trained CNN architecture in terms of its efficiency and accuracy performance when diagnosing pneumonia. First of all, this is an issue because:

1. Accuracy is not the only measure worth considering, as a model achieving a certain level of accuracy would be impossible to use for making predictions if it required huge amounts of memory and time.
2. Models deployed in different environments have distinct requirements. For instance, deploying a model onto a hospital computer is different from deploying a model to a mobile phone or edge device in the field.
3. There is no comparison in terms of efficiency provided in the majority of studies (Nguyen & Hy, 2026).

Thus, the research question being studied is: Which pre-trained CNN architecture can be selected for pneumonia prediction based on both accuracy and computational efficiency in low-resource settings?

1.4 Purpose of the Research

The goal of this study is to compare and evaluate the effectiveness of using VGG16, ResNet50, and EfficientNet to detect pneumonia using chest X-ray images, taking into consideration not only their accuracy but also their efficiency, and make recommendations accordingly.

1.5 Theoretical Framework

This research is grounded in two theoretical domains:

Theory of Transfer Learning: Knowledge learned during a specific task (e.g., ImageNet classification) can be applied to other similar tasks (e.g., medical image classification) using the

learned features representations (Ahmad et al., 2025). This approach eliminates the need for collecting a large number of labelled medical images.

Multi-Objective Optimization: The notion that real-life systems must consider different objectives simultaneously, such as maximizing the performance level while reducing the amount of resources needed (Nguyen & Hy, 2026). The ideal solution is determined by how much weight is placed on each objective.

1.6 Significance of the Research

This research is significant for three reasons:

1. **Scholarly Contribution:** It fills an identified gap in the existing literature through the provision of the first comparison of the two parameters in pneumonia diagnosis using the three types of CNN architecture (Asnaoui et al., 2020).
2. **Practical Contribution:** The paper offers useful guidelines for healthcare professionals, technology developers, and policy-makers in choosing appropriate CNN architectures to be used in resource-constrained environments (Nguyen & Hy, 2026).
3. **Methodological Contribution:** The study presents a replicable method of assessing artificial intelligence models not only in terms of their performance but also in terms of their efficiency (Ahmad et al., 2025).

1.7 Feasibility, Scope and Limitations

1.7.1 Feasibility

This study can be conducted in 12 weeks with the help of an ordinary personal computer. All software needed for the study (Python, PyTorch, and Scikit-learn) is open-source and freely available.

1.7.2 Scope

The research involves binary classification (Normal vs. Pneumonia) using the Chest X-ray Pneumonia data set (Mabrouk et al., 2024). Multi-class classification (for example, bacterial and viral types of pneumonia), and other diseases are not covered in the research.

1.7.3 Limitations

- Results are not necessarily transferable to other imaging tasks or different chest X-ray data sets
- Efficiency measures vary by computing hardware used (specifications will be included with results)
- Clinical validation of the outputs from the machine learning models is not included

1.8 Research Aims and Objectives

1.8.1 Aim

The purpose of this study is to assess and contrast the performance of VGG16, ResNet50, and EfficientNet based on accuracy and computational efficiency when used for pneumonia classification using chest X-rays, and ultimately determine the best model.

1.8.2 Objectives

1. To implement transfer learning using pre-trained VGG16, ResNet50, and EfficientNet architectures for binary classification of chest X-rays (Normal vs. Pneumonia)
2. To evaluate the classification accuracy of each architecture using standard metrics including precision, recall, F1-score, and area under the ROC curve (AUC-ROC)
3. To measure the computational efficiency of each architecture including model file size (MB), inference time per image (milliseconds), and peak memory usage during inference (MB)
4. To compare results across all three architectures and identify which model offers the optimal balance between accuracy and efficiency
5. To produce a comprehensive recommendation for model selection based on different deployment scenarios (maximum accuracy vs. maximum efficiency vs. balanced approach)

1.9 Preliminary Literature Review

1.9.1 Existing Research in the Domain

There has been increasing attention paid to applying deep learning methods to interpreting X-rays of the chest since 2017. In particular, a study named CheXNet showed that a CNN architecture of 121 layers was capable of achieving performance superior to that of practicing radiologists in detecting pneumonia (Rajpurkar et al., 2017). Further evidence supporting this conclusion has come from follow-up research (Ahmad et al., 2025).

Several comparative studies on CNN architectures have produced conflicting results. For example, while some researchers claim that deeper architectures such as ResNet are superior to shallower alternatives like VGG (Dharmik, 2025), others argue that the choice of architecture is not as important as how well the model is trained (Asnaoui et al., 2020). EfficientNet, a recently proposed architecture, has proven to perform excellently on natural images but has yet to be widely explored within medicine (Filvantorkaman & Torkaman, 2025).

1.9.2 Strengths, Weaknesses, Challenges and Opportunities

1.9.3 Existing strengths:

- Consensus on capability to achieve high accuracy on the task (Rajpurkar et al., 2017).
- Established approach for transfer learning in medical imaging (Ahmad et al., 2025).
- Availability of benchmark datasets publicly (Mabrouk et al., 2024).

1.9.4 Existing weaknesses/gaps:

- Few studies consider efficiency of computation (Nguyen & Hy, 2026).
- Absence of any comparative study on VGG16, ResNet50 and EfficientNet with respect to both factors (Asnaoui et al., 2020).
- No suggestions on how to choose among the models in case of limited resources

1.9.5 Existing challenges:

- Use of various splits between training and testing datasets.
- Differing hardware between studies affects efficiency assessment.

- Potential dataset shift when comparing results from various sets of chest x-rays.

1.9.6 Existing opportunities:

- Need to establish a common framework for comparative purposes (Nguyen & Hy, 2026).
- Increased interest in efficient computing due to growing interest in edge devices (Filvantorkaman & Torkaman, 2025).

1.9.7 Justification of Proposed Research

The need for such a study stems from the above-mentioned gap – the lack of an existing comparison between these three types of neural network architectures in terms of their accuracy and efficiency (Asnaoui et al., 2020). Moreover, the problem is of great practical importance due to its impact on providing medical services in underdeveloped regions (Nguyen & Hy, 2026). Doctors and engineers do not have any data-driven guidelines as to which CNN architecture should be employed.

1.9.8 The Gap to be filled

The research problem that this study seeks to solve is:

There is no previous work that has compared the performance of VGG16, ResNet50, and EfficientNet models for detecting pneumonia while considering accuracy and computational efficiency simultaneously (Asnaoui et al., 2020);(Nguyen & Hy, 2026).

The specific gap this research addresses is: There is no prior study that has ever compared all three models, namely VGG16, ResNet50, and EfficientNet, at once in terms of their performance in detecting pneumonia along with assessing their computing cost effectiveness.

1.10 Experimental Design and Methods



Figure 1: Sample chest X-ray images (Normal vs. Pneumonia)

1.10.1 Research Approach

The experiment is designed using a quantitative comparative methodology. There will be three independent variables (the CNN models). These will be measured against two dependent variable sets: accuracy measures and efficiency measures. The other factors involved will be kept constant for fairness of the comparisons.

1.10.2 Dataset

The Chest X-Ray Images (Pneumonia) dataset, available on Kaggle, will be used. The dataset contains 5,863 chest X-ray images classified into two categories (Mabrouk et al., 2024):

CLASS	NUMBER OF IMAGES
NORMAL	1341
PNEUMONIA	3875
TOTAL	5216

Table 1.1 : Dataset class distribution

The 'Chest X-ray Images (Pneumonia)' dataset consists of 5,863 image samples divided into two classes, as displayed in Table 1. In Table 1, one can observe that there is an imbalance between the numbers of images in each category where the number of pneumonia images (4,280) is more than that of normal images (1,583) in the ratio of about 2.7:1. This will be remedied during the training process using stratified sampling and employing a weighted loss function to avoid favoring the dominant class.

The dataset is publicly available under an open license, has been validated by medical experts, and is widely used as a benchmark in existing literature (Asnaoui et al., 2020). No ethical approval is required as the data is de-identified and publicly released.

1.11 Sequence of Procedures

1.11.1 Phase 1: Data Preparation (Week 3)

1. Download dataset from Kaggle (Mabrouk et al., 2024).
2. Organise images into directory structure (train/val/test)
3. Resize all images to 224×224 pixels
4. Normalise pixel values to [0, 1]
5. Split data: 70% training, 15% validation, 15% testing (stratified to maintain class balance)
6. Apply data augmentation to training set only: random rotation ($\pm 10^\circ$), horizontal flip, zoom ($\pm 10\%$)

1.11.2 Phase 2: Model Implementation (Weeks 4-6)

1. Load pre-trained VGG16, ResNet50, and EfficientNet-B0 from torchvision (He et al., 2016; Simonyan & Zisserman, 2015; Tan & Le, 2020).
2. Replace final classification layer with binary output (1 neuron, sigmoid activation)
3. Freeze early layers (first 80% of layers)
4. Configure training with identical hyperparameters for all models:

HYPERPARAMETER	VALUE
Optimizer	Adam
Learning rate	0.0001
Batch size	32
Loss function	Binary cross-entropy
Epochs	20 (with early stopping, patience=5)

Table 1.2: *Training hyperparameters*

Table 2 contains the list of hyperparameters that were used during the training of all three models. The Adam algorithm was chosen because of its adaptive learning rate capability. An adaptive learning rate is particularly important when performing the task of classifying medical images. The standard learning rate in fine-tuning pre-trained models is 0.0001.

1.11.3 Phase 3: Training and Evaluation (Weeks 4-7)

1. Train each model sequentially, recording training history
2. After training, evaluate on held-out test set
3. Generate confusion matrices and ROC curves for each model

1.11.4 Phase 4: Efficiency Measurement (Week 8)

1. Save each trained model and measure file size (MB)
2. Measure inference time: average over 1,000 test images (10 runs) (Filvantorkaman & Torkaman, 2025).
3. Measure peak memory usage during inference using psutil library

1.11.5 Phase 5: Analysis and Recommendation (Week 9)

1. Compile results into comparison table
2. Identify which model is optimal for each scenario (Nguyen & Hy, 2026):
 - Maximum accuracy (no efficiency constraint)
 - Maximum efficiency (minimal hardware)
 - Balanced approach (recommended for typical low-resource settings)

1.12 Data Analysis and Interpretation

1.12.1 Accuracy Analysis:

- Calculate precision, recall, F1-score, and AUC-ROC for each model (Dharmik, 2025).

- Compare using standard classification report format
- Statistical significance testing (McNemar's test) to compare models

1.12.2 Efficiency Analysis:

- Tabulate model size (MB), inference time (ms), memory usage (MB)
- Create efficiency ranking

1.12.3 Trade-off Analysis:

- Compute composite score = (normalised accuracy) + (normalised inverse efficiency) (Nguyen & Hy, 2026)
- Identify highest composite score as recommended model

1.13 Potential Challenges and Mitigation Strategies

CHALLENGE	MITIGATION STRATEGY
Laptop insufficient for training all three models	Use Google Colab free GPU; reduce batch size; use fewer epochs; prioritize two models
Dataset download issues	Alternative identified: NIH Chest X-ray dataset
Models overfit to training data	Use dropout (p=0.5), data augmentation, early stopping with validation monitoring
Inconclusive or unexpected results	Report as valid finding; discuss possible explanations; suggest future work
Time constraints	Strict weekly schedule; complete one model before starting next; prioritize writing

1.14 Chapter Summary and Transition

This chapter introduced the critical challenge of diagnosing pneumonia in low-resource clinical settings and outlined the necessity of balancing classification accuracy with computational efficiency when selecting deep learning models. It established the primary aim, objectives, theoretical frameworks, and scope of this dissertation to address the clear gap in existing literature regarding a simultaneous comparison of VGG16, ResNet50, and EfficientNet-B0. To ground these objectives within the broader academic discourse, Chapter 2 provides a comprehensive review of the literature on medical image analysis, the evolution of deep learning in radiology, and the specific architectural trade-offs identified in prior research.

2. : Literature Review

2.1 Introduction

This chapter reviews existing literature on medical image analysis using deep learning, focusing specifically on chest X-rays for pneumonia detection. The review will explore current state of research in the field of deep learning, providing comprehensive understanding of trends, identifying gaps in literature and justifying the need for the study. The chapter will begin by examining the global burden of pneumonia and how automated X-rays analysis helps to diagnosis. Building upon this discussion, it then traces how deep learning evolved in medical imaging, from early convolutional neural networks (CNNs) to current dominance of transfer learning. Expanding on this foundation, the three CNN architectures used for this study: VGG16, ResNet50 and EfficientNet are examined in detail. With these core principles set out, the write up will discuss strengths and weakness in this comparative study, highlighting opportunities for future work. Then finally, examine the literature to establish a research gap that this dissertation addresses.

2.2 Pneumonia and Chest X-ray Diagnosis

2.2.1 Global burden of pneumonia

According to the World Health Organization (2022), pneumonia is a form of acute respiratory infection that affects the lungs. Inside the lungs are microscopic air sacs called alveoli, which fill with air when a healthy person breathes; however, when an individual has pneumonia, the alveoli fill with pus and fluid, making breathing painful and limiting oxygen intake. The disease is caused by several micro-organisms, including viruses, bacteria, and fungi.

Pneumonia remains the single largest cause of mortality among children worldwide. In 2019 alone, it killed 740,180 children under the age of 5 accounting for about 14% of all deaths for children under 5 and 22% of deaths for children between 1 and 5 years old with the highest records of deaths occurring in southern Asia and Sub-Saharan Africa. Nevertheless, the disease can be prevented with immunization against *Haemophilus influenza* type b (Hib), pneumococcus, measles, and whooping cough, and it can be treated effectively with low-cost antibiotics like amoxicillin (World Health Organization, 2022).

2.2.2 Chest X-ray as a Diagnostic Tool

Chest X-rays to date is the main diagnostic tool used for pneumonia detection. However these chest X-rays require specialized training and vast amount of experience. Distinguishing between viral or bacterial pneumonia often cause disagreements amongst radiologist , resulting in diagnostic variability (Rajpurkar et al., 2017). In low resource clinical settings such as developing countries , often under staffed with few or no radiologist , computer imaging diagnostics tools provides a strong panacea and justification for research into automated low cost pneumonia detection systems .

2.3 Deep Learning in Medical Imaging: Evolution and Key Milestones

2.3.1 Early Convolutional Neural Networks

Convolutional neural networks (CNNs) have been applied in medical image analysis since their inception, with early medical applications dating back to the late 1980s and 1990s (Litjens et al., 2017). Primitive CNN architectures, most notably AlexNet, proved that deep learning could achieve groundbreaking performance on natural image classification tasks(Litjens et al., 2017). However, early adoption was significantly constrained by the scarcity of large, expert-annotated datasets and limited computational resources (Litjens et al., 2017; Shen et al., 2017).

When large-scale medical imaging datasets , such as ChestX-ray14 was released , containing over 100 000 chest x-ray images labelled with up to 14 diseases , breakthroughs into deep learning exploded for medical image diagnosis (Rajpurkar et al., 2017). This labelled dataset provided foundations for further research in the area of CNN-based medical algorithms.

2.3.2 CheXNet: A Breakthrough in Pneumonia Detection

A crucial study in the application of deep learning in chest X-ray analysis came with the development of CheXnet (Rajpurkar et al., 2017). The neural network CheXnet is a 121-layer CNN trained on the ChestX-ray14 dataset. The study was conducted by four practicing academic radiologists who labelled the test set, which was used to compare the performance of a radiologist and model on the same dataset. The researchers discovered that CheXnet diagnosis performed better than actual human radiologist on both sensitivity and specificity.

This result was paramount in many aspects. First it proved that automated deep learning models can be used to diagnose medical imaging and even performing better than practicing human

professionals. Secondly it demonstrated that models trained on large dataset could generalize well to radiologist-annotated test set. Thirdly this also showed clinical radiologist can also work hand-in-hand with AI systems to provide accurate diagnosis. Ultimately the researchers developed a heat-map tool (Grad-CAM) used to visualize areas on the image X-ray where the algorithm determined were likely to contain pneumonia, which speed radiologist workflow (Rajpurkar et al., 2017).

The success of CheXnet gave birth to new field in deep learning , named transfer learning , where trained models are fine-tuned on different large image dataset with less iteration than starting completely from scratch.

2.3.3 Transfer Learning in Medical Imaging

Ahmad et al. (2025) performed a comprehensive study of transfer learning application in medical image classification using pre-trained deep learning CNN models. The analysis evaluated six pre-trained models (AleNet, InceptionV3, ResNet18, ResNet34, ResNet50 and VGG16) on a labelled chest X-ray image dataset. The results showed InceptionV3 performed better other five models, followed by ResNet family models, VGG16 and AlexNet performed well but with low accuracy.

Ahmed et al. (2025) also compared runtime comparison, to gauge robustness and computational efficiency on the six models, they found out that pre-trained models are beneficial in limited computation resources, as much of the model training would have been done when the models were initially created, thus few epoch will be required for transfer learning. Improvement on the models depended on several factors such as, image domain similarity (Chest X-ray medical images on models pre-trained on Chest X-ray images), model architecture and dataset size. Theses factor were recommend for efficient future predictions. These findings are relevant to the current study, which focuses on evaluating trade-off between accuracy and efficiency.

2.4 CNN Architectures for Pneumonia Detection

2.4.1 VGG16

VGG16 has a 16-layer CNN in its architecture and depth (13 convolutional layers and 3 fully connected layers) (Simonyan & Zisserman, 2015). VGG16's uniform architecture, small 3x3 filters throughout makes it conceptually simple and easy to implement, but its large parameter count of 138 million results in large model size and relatively slow inference.

El Asnaoui et al. (2020) compared modern deep architectures for automatic binary classification of pneumonia images, using fine-tuning versions of VGG16, VGG19, DenseNet201, Inception_ResNet_V2, Inception_V3, ResNet50, MobileNet_V2 and Xception. The study utilized X-Ray images dataset containing 5 856 images, split (4 273 Pneumonia, 1583 normal). The results concluded that fine-tuned versions of MobileNet_V2 , ResNet50 and Inception_V3, ResNet50 worked effectively in both training and validation with accuracy surpassing 96% . On the other end models such as VGG16, VGG19, DenseNet201and Inception_V3 displayed lower performance, accuracy averaging 84%.

Transfer learning approaches have also been evaluated on their effectiveness in diagnosing COVID-19 pneumonia, Dharmik (2025) evaluated (ResNet50, ConvNeXtTiny, MobileNet, NASNetMobile and DenseNet121 using chest X-Ray and CT Images. DenseNet121 emerged as the best performing architecture with an accuracy of 98%, Precision of 96.9, a recall of 98.9% and F1 Score of 97.9%. The confusion matrix showed minimal false positives and false negatives, highlighting the model's applicability in real-world diagnostics scenarios. The study further justifies the effectiveness of pre-trained models for pneumonia detection, however optimal performance is highly dependent on dataset used, model architecture and the task complexity.

2.4.2 ResNet50

This model introduced residual connections or skip node connections , to allow gradients to flow directly through the network (He et al., 2016). The architecture was proposed to mitigate the vanishing gradient problem, enabling the training of much deeper networks without degrading performance. ResNet50 is built on 50 layers and approximately 25 million parameters.

The residual connections mechanism in ResNet50 allows it to learn identity mappings, thus if a particular layer fails to improve performance, the network effectively skips it. This property of ResNet50 makes it well suited for classification task where patterns must be detected, such as medical image analysis. Whilst comparing pre-trained models El Asnaoui. (2020) identified ResNet50 as one of the top-performing models, with accuracy exceeding 96%.

However a recent study on breast tumor detection using CT scans imaging found out that VGG16 outperformed deeper architectures including ResNet and EfficientNet (Alzubaidi, et al., 2021). They concluded that modern deep CNN architectures , which includes ResNet52V2 and EfficientNetB5 portray poor generalization when fine-tuning on medium - sized medical image

dataset , while simpler architectures like VGG16 demonstrates superiority. The counter-intuitive finding points to the importance of architecture selection relative to dataset size.

2.4.3 EfficientNet

EfficientNet makes use of compound scaling to balance network depth , width and resolution (Tan & Le, 2020). Contrary to most architectures discussed previously, EfficientNet applies a systematic scaling method that uniformly scales all dimensions using a fixed set of coefficients, whereas other architectures scaled individual dimensions arbitrarily. Consequently this mechanism produce models that achieves high accuracy whilst utilizing fewer parameters and less computational resources.

EfficientNet-B0 is the smallest of the family of models, utilizing around 5.3 million parameters, thus its attractiveness for deployment in resource constrained settings such as edge device and mobile platforms. Despite its effectiveness , EfficientNet is not yet fully explored in medical field (Tan & Le, 2020). (Filvantorkaman & Torkaman, 2025) conducted a study on real-time medical image analysis, using deep neural networks pre-trained architectures like U-Net, EfficientNet and Transformer- based models, with real-time optimization strategies. The used pruning, quantization and GPU acceleration. This framework, inclusive of EfficientNet attained accuracies above 92% and lower inference times of 80 milliseconds, demonstrating the potential of EfficientNet for real-time clinical application.

2.5 Accuracy versus Efficiency: The Deployment Challenge

2.5.1 The Efficiency Challenge

A crucial issue to consider when deploying deep learning models in healthcare is the gap between theoretical efficiency (measured by FLOPs and parameter count) and actual performance on healthcare hardware. Nguyen and Hy (2026) have done a study to address the comprehensive synthesis of efficient lightweight deep learning models in medical domain. They argued FLOPs and parameter count are not the only metrics to be used for measuring efficiency, because real-world clinical deployment particularly on edge devices such as mobile NPU, rely more on Memory Access Cost (MAC) and parallelism.

This mismatch leads to what Nguyen and Hy (2026) have defined as the efficiency gap: Architectures tuned for maximum FLOPs alone tend to exhibit inefficient hardware usage because

of too many fragmentations of depth wise convolution or sparsity of connections, which means that these systems have inference speeds that do not match their theoretical lightweight nature. Thus, the current measure of deployment readiness has moved away from being FLOP focused optimization to the level of system optimization and considerations like memory contiguity, operator fusion compatibility, and dimension consistency.

It should be noted that Nguyen and Hy (2026) found out that practical optimization of efficiency usually requires the implementation of the execution framework such as TensorRT, ONNX Runtime, OpenVINO, CoreML, NNAPI, and TVM, that converts a pre-trained model into optimized kernels for hardware execution. Efficiency of architecture does not automatically equal practical speedup without runtime and compiler infrastructure support. This particular finding is of great importance for the current study because of its focus on practical advice in terms of low-resource healthcare settings.

2.5.2 Efficiency-Aware Reviews

Surveys that have examined edge deep learning and light weight models have looked at compression approaches and hardware limitations in the context of point of care devices and mobile devices, stressing latency, memory, and energy consumption. However, the evaluation in such surveys is quite general, and does not involve creating a framework for comparison of entirely different architectures in a consistent clinical context. There are also not many empirical studies that examine the behavior of such models in the real world in terms of latency, memory, and energy.

The above-mentioned shortcomings in the current literature necessitate empirical research that compares different architectures not only in terms of accuracy but also in terms of efficiency on the same hardware. The current research helps fill this gap by performing an analysis of VGG16, ResNet50, and EfficientNet in terms of accuracy and efficiency, where efficiency is defined in terms of CPU inference time and memory usage.

2.6 Strength, Weaknesses, Challenges and Opportunities in Existing Research

2.6.1 Strength

The literature from past studies demonstrate applicability of deep learning models on medical image diagnosis , achieving high accuracy on detecting pneumonia from chest X-rays images

(Dharmik, 2025; Rajpurkar et al., 2017). Deep learning models such as CheXNet with accuracies above 95%, proved CNNs models can surpass practicing Radiologist accuracy which ranges between 90-95% (Rajpurkar et al., 2017).

Pre-trained models have been reused on a different datasets , fine tuning them to suit context (transfer learning technique), worked exceptionally well on medical imaging (Ahmad et al., 2025). They worked on ImageNet dataset, to establish transfer learning concept, despite limited medical imaging data. Other dataset such as ChestXray 14 and Kaggle pneumonia dataset have been developed, to provide benchmarking in medical imaging , thus researchers have plenty of option to choose upon (Mabrouk et al., 2024).

2.6.2 Weaknesses

Though literature suggest many benefits of deploying deep learning and transfer learning techniques on medical diagnosis, they mainly focus on accuracy of prediction , without considering computational efficiency (Nguyen & Hy, 2026). This current study will explore that gap, to see how chosen models perform measuring accuracy and efficiency. While some studies have been done to measure accuracy and efficiency on individual models ((Asnaoui et al., 2020) none has been done to compare VGG16, ResNet50 and EfficientNet with respect to accuracy and efficiency. This gap makes it difficult for practitioners to select the best model for their given context. Furthermore there is no guidance on how to choose amongst models, when in limited resource settings, particularly computational resources like RAM, Storage of CPU.

2.6.3 Challenges

The literature points to a cluster of methodologies, without one agreeable flow. Different train-test splits makes it difficult to compare results, even when the same dataset is used. These anomalies in splitting data leads to different reported accuracies (Nguyen & Hy, 2026) .

There is no uniformity in hardware specifications applied in most of the studies, thus one model may appear to be slow in one study and fast on other studies, due to differences in CPU, GPU or optimization framework used. Lastly models trained on different dataset from one institution or particular demographic area may not generalize of other settings, making cross-study comparison difficult.

2.6.4 Opportunities

Previous study show several opportunities for future research, chiefly being the need for establishing framework for benchmarking studies (Nguyen & Hy, 2026). This framework will standardize evaluation metrics such as accuracy and efficiency to facilitate comparison across studies.

Literature suggest a growing need for high computing power to process deep learning networks (Filvantorkaman & Torkaman, 2025) and also an ever-growing usage of edge and mobile devices creates demands for lightweight models which can run on less computing resources.

2.7 Justification of the proposed research

The necessity for this research comes from the gap found in the literature, the absence of comparison between VGG16, ResNet50 and EfficientNet regarding their accuracy and efficiency (El Asnaoui et al., 2020). Furthermore, the importance of the issue can be seen in its application for the delivery of medical services in less developed areas (Nguyen & Hy, 2026). There are no data-driven guidelines for doctors and engineers to use in order to understand which architecture of CNN to use in case of limited computational resources.

This research is able to fill this gap through presenting the first simultaneous comparison of all three architectures regarding accuracy and efficiency.

2.8 The Gap to be filled

The research problem that this study seeks to solve is:

There is no previous work that has compared the performance of VGG16, ResNet50, and EfficientNet models for detecting pneumonia while considering accuracy and computational efficiency simultaneously (El Asnaoui et al., 2020; Nguyen & Hy, 2026).

The specific gap this research addresses is:

No prior study has compared all three models VGG16, ResNet50, and EfficientNet in terms of their performance in detecting pneumonia along with assessing their computing cost-effectiveness.

2.9 Chapter Summary and Transition

This chapter has reviewed the existing literature on deep learning approaches for pneumonia detection from chest X-rays. Key findings include:

- Pneumonia is a major global health burden, and chest X-ray remains the primary diagnostic tool.
- CheXNet (Rajpurkar et al., 2017) established that CNNs could surpass radiologist-level performance, catalyzing research in this domain.
- Transfer teaching (Ahmad et al., 2025) has become the standard approach for medical image classification, with InceptionV3 and ResNet50 consistently performing well.
- VGG16, ResNet50, and EfficientNet each offer different trade-offs between accuracy and efficiency, but no study has directly compared all three on both dimensions under identical conditions (El Asnaoui et al., 2020; Nguyen & Hy, 2026).
- Efficiency metrics (model size, inference time, memory usage) are rarely reported, yet they are critical for deployment in resource-constrained settings.
- The present study addresses this gap by providing a direct comparison of all three architectures on accuracy and efficiency, offering practical guidance for model selection in low-resource healthcare settings.

The next chapter (Methodology) describes the experimental design, dataset, models, training protocol, and evaluation metrics used to address the research gap identified in this review.

3. Methodology

3.1 Introduction

This research aims to check how different Convolutional Neural Network designs perform under the same conditions, focusing on their trade-offs. The chapter lays out the plan, chosen network architectures, how performance will be measured, and the controls for a thorough, repeatable comparison.

3.2 Research Approach and Experimental Design

This study uses a quantitative, comparative experiment. It focuses on looking at how different structures in deep learning models affect prediction accuracy and computational costs. For the research, I picked three basic CNN designs that show various approaches in the development of deep learning: VGG16, which is deep and sequential; ResNet50, using residual learning and skip connections; and EfficientNet-B0, known for its compound scaling method.

Independent Variable: The structural architecture of the convolutional neural network (VGG16, ResNet50, EfficientNet-B0).

Dependent Variables: The performance outcomes, bifurcated into two primary dimensions:

Classification Accuracy: Quantified via macro-averaged Accuracy, Precision, Recall, F1 Score, Area Under the Receiver Operating Characteristic curve (AUC-ROC), and a multi-class Confusion Matrix.

Computational Efficiency: Quantified via Model File Size (MB), CPU Inference Latency (ms/image), and Peak Memory Consumption (MB).

3.2.1 Experimental Controls and Standardization

To guarantee internal validity and isolate the architecture as the sole differentiator of performance, all confounding variables were strictly controlled. The experimental pipeline enforces uniformity across the following dimensions:

Experimental Domain	Controlled Parameter Standard
Data Environment	Identical dataset splits (Training, Validation, and Testing) across all models.

Pipeline Processing	Standardized image resizing, normalization vectoring, and deterministic data augmentation.
Optimization	Identical training hyperparameters, including learning rate, optimizer (e.g., Adam/SGD), batch size, and loss function.
Hardware Environment	Execution of all inference and evaluation pipelines on a dedicated, identical hardware configuration to ensure CPU latency and peak memory metrics are directly comparable.

Table 3.1: Experimental Controls and Standardization Baseline

3.3 Dataset Description

3.3.1 Source, Origin, and Clinical Context

This study uses the well-known Chest X-Ray Images (Pneumonia) dataset from Kermany et al. (2018), which others have also analyzed (like Mabrouk et al., 2024). You can find it on Kaggle, and it’s available under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. This lets researchers like me use it for our studies and compare results too.

The dataset's clinical and physical attributes are structured as follows:

Clinical Source: Guangzhou Women and Children’s Medical Center, Guangzhou, China.

Patient Cohort: Pediatric patients aged between one and five years. This specific demographic is highly critical, as pneumonia remains a leading cause of mortality in children under five globally.

Imaging Modality: Digital Anterior-Posterior (AP) chest radiographs. AP views are standard in pediatric imaging but present unique challenges due to patient movement and anatomical variations.

Format and Volume: The raw dataset comprises 5,216 digital images stored in standard joint photographic experts group (JPEG) format.

3.3.2 Annotation, Ethical Standards, and Class Distribution

To set up a solid clinical gold standard, the dataset went through several checks by experts before being published. First, two physicians did the initial review and annotations. When there was a disagreement, a more experienced radiologist stepped in to help reach an agreement. This extra step lowers bias and makes the diagnostics reliable.

The dataset shows a class imbalance split into two main groups, though:

NORMAL Class (N = 1,341): Depicts clear lungs without any clear areas of opacification.

PNEUMONIA Class (N = 3,875): Manifests as focal or diffuse opacities. This class captures both bacterial pneumonia (typically exhibiting focal lobar consolidation) and viral pneumonia (typically exhibiting diffuse interstitial patterns).

Total Dataset Volume (N = 5,216):

NORMAL Class: 1,341 images (~25.71%)

PNEUMONIA Class: 3,875 images (~74.29%)

Ethical Statement: The data was completely anonymized before being released to the public, with all patient info removed. Consequently, following standard IRB rules, I didn't need explicit ethical approval or informed consent for my secondary analysis.

3.3.3 Data Cleaning, Quality Assurance, and Preprocessing

To address the supervisor feedback requirement for an explicit data cleaning narrative, the raw images underwent a rigorous quality assurance and preprocessing pipeline prior to being fed into the VGG16, ResNet50, and EfficientNet architectures:

Artifact Removal and Quality Filtering: My data had some corrupt file headers, unreadable images, and low-quality radiographs. I used a Python script with Pillow to check file integrity. Images with major motion artifacts or severe pixel clipping were automatically removed to keep the models from learning clinical irrelevant stuff.

Dimension Standardization (Resizing): The raw X-ray images possessed highly variable spatial resolutions (ranging from 400 x 400 to over 2000 x 2000 pixels). Because deep

CNNs require fixed-size tensor inputs, all images were downsampled using bilinear interpolation to a uniform dimension matching the native inputs of the target architectures:

VGG16 & ResNet50: Resized to 224 x 224 x 3 pixels.

EfficientNet-B0: Resized to 224 x 224 x 3 pixels.

Intensity Normalization: To neutralize variations in X-ray exposure, contrast, and machine calibration, global contrast normalization was applied. Pixel intensities, originally bounded in $[0, 255]$, were rescaled to a floating-point range of $[0.0, 1.0]$. Subsequently, Z-score normalization was performed using the standard ImageNet channel means ($\mu = [0.485, 0.456, 0.406]$) and standard deviations ($\sigma = [0.229, 0.224, 0.225]$) to stabilize gradient descent during training.

3.3.4 Relevance to the Research Problem and Evaluation Alignment

The structural characteristics of this dataset align precisely with the comparative objectives of this dissertation:

Real-World Clinical Simulation: The dataset presents a highly realistic diagnostic environment. Pediatric chest radiographs are notoriously noisy, featuring variations in lung inflation and patient positioning, which thoroughly tests the generalization capabilities of the three CNN architectures.

Imbalance Handling Efficacy: The 74.29% to 25.71% tilt towards the pneumonia class matches what I see in real-world settings. So, this skew supports the evaluation matrix set up in Section 3.1. If I only use overall accuracy, I'd get misled, so my study uses specific metrics like Precision, Recall, F1-Score, and AUC-ROC. These ensure that each system is fairly evaluated on how it deals with the smaller class under the same conditions.

Benchmark Standardization: Since this dataset is commonly used in current medical AI studies, using it lets the dissertation compare its results with existing standards. This way, both how well the computations work and how accurate the classifications are can be clearly compared to past work.

3.4 Data Cleaning and Quality Assurance Pipeline

To ensure my study's results are reliable and avoid any hidden errors, I ran a thorough data cleaning process. This is key for comparing deep learning models since issues like label mistakes can really mess up the performance metrics and give me misleading results.

3.4.1 Phase 1: Format and Integrity Verification

The initial phase programmatically parsed all 5,216 raw files to verify data types and corrupted payloads:

MIME/Extension Filtering: The pipeline enforced a strict structural whitelist, retaining only standard compressed graphic formats (.jpeg, .jpg, and .png). Any auxiliary metadata or non-standard system files were automatically filtered.

Corrupted Byte Detection: To safeguard the training pipeline against runtime computational faults, every image was lazily loaded using the Python Imaging Library (PIL.Image.open ()). The execution block monitored for UnidentifiedImageError exceptions.

Phase Empirical Outcome: All 5,216 source files successfully passed byte-level instantiation, confirming zero file corruption in the primary repository.

3.4.2 Phase 2: Structural Directory Validation

The dataset's underlying storage architecture was parsed to ensure adherence to a standard machine learning directory topology. The pipeline verified the absolute paths of the target splits and ground-truth subdirectories:

```
chest_xray/
├──train/
│   ├──NORMAL/
│   └──PNEUMONIA/
├──val/
│   ├──NORMAL/
│   └──PNEUMONIA/
└──test/
```

|—NORMAL/
|—PNEUMONIA/

Pipeline Invariant: The initialization script was bound to a hard failure condition. If any component of this hierarchical tree structure was missing or malformed, the pipeline terminated execution immediately with an explicit path-error exception rather than proceeding with partial data.

3.4.3 Phase 3: Categorical Consistency and Expert Spot-Checking

Rather than relying on brittle, highly variable string patterns within filenames, categorical labels were deterministically mapped using the immediate parent directory name:

Parent Folder = "NORMAL" -> Class Label = 0

Parent Folder = "PNEUMONIA" -> Class Label = 1

To check the accuracy of the source annotations and catch any mistakes, a random spot-check was done. From each class, fifty radiographs were picked and looked over. They checked for visual signs, like clear lung fields in Class 0 and visible problems in Class 1. This manual process confirmed everything matched up perfectly with the folder designations, so I know the data is reliable.

3.4.4 Phase 4: Cryptographic Duplicate Detection

Data leakage from identical images appearing in both training and evaluation splits can hugely inflate model performance metrics, making the model seem way more accurate than it is. To stop this, I used MD5 cryptographic hashing to create unique, 128-bit digital fingerprints for every image. For exact duplicates, the pipeline checked all the hashes and didn't find any identical files across different classes or splits. For perceptual duplicates: those with minor pixel shifts or compression differences but the same clinical content, I kept them on purpose. This lets the models deal with the realistic variations I see in digital radiography calibration.

3.4.5 Phase 5: Automated Statistical Anomaly and Image Quality Filtering

To isolate low-quality radiographs that could destabilize gradient updates during backpropagation, pixel-level descriptive statistics were computed across the dataset. Images were flagged for automated drop-out based on mathematical thresholds:

Under-exposure Threshold: Mean Pixel Intensity (μ) < 10 (indicative of blank or heavily clipped black artifacts).

Over-exposure Threshold: Mean Pixel Intensity (μ) > 245 (indicative of washed-out white saturation).

Information Entropy Threshold: Standard Deviation (σ) < 5 (indicative of flat, completely uniform matrices devoid of diagnostic features).

Following programmatic analysis across the 5,216 files, no images violated these boundaries, indicating a highly uniform collection baseline.

3.4.6 Phase 6: Stratified Dataset Partitioning

Following verification and cleaning, the dataset was partitioned using a strict 70 / 15 / 15 stratified split for training, validation, and testing configurations, respectively. Applying a stratified split is vital given the dataset's inherent class imbalance; it guarantees that the 74.29% to 25.71% diagnostic ratio is perfectly preserved across all three operational subsets, neutralizing statistical variance during evaluation.

Experimental Split	NORMAL (Class 0)	PNEUMONIA (Class 1)	Total Volume	Class Imbalance Ratio
Training	939	2,713	3,652	74.29% / 25.71%
Validation	201	582	783	74.33% / 25.67%
Testing	202	582	784	74.23% / 25.77%
Aggregate Total	1,342	3,877	5,219	74.29% / 25.71%

Table 3.2: Final Stratified Dataset Distribution

Methodological Wall: To guarantee absolute empirical validity and avoid data snooping biases, the test partition ($N = 784$) was completely sequestered during optimization. It was never accessed during hyperparameter tuning, feature extraction, or model validation checkpoints, and was introduced exclusively during the final benchmark phase.

3.5 Data Preprocessing and Augmentation Pipeline

To optimize architectural convergence and minimize structural overfitting, the dataset underwent a bifurcated pipeline consisting of deterministic preprocessing and stochastic data augmentation. While preprocessing transformations were applied globally to maintain consistency across all subsets, data augmentation was restricted exclusively to the training partition to serve as an explicit regularisation mechanism.

3.5.1 Mandatory Preprocessing (Training, Validation, and Testing Sets)

To address the restrictions imposed by the input tensor sizes of the VGG16, ResNet50, and EfficientNet-B0 models, all test images underwent a consistent preprocessing process. This will allow the testing of accuracy and speed without being affected by changes in image dimensions.

Transformation Operation		Technical Specification	Methodological Rationale
Spatial Resampling (Resize)		224 x 224 pixels via Bilinear Interpolation	Harmonizes the variable raw radiograph dimensions to standard square matrices matching the native receptive fields of the target CNN architectures.
Tensor (ToTensor)	Conversion	Scaling: [0, 255] to [0.0, 1.0]	Casts the 8-bit integer pixel intensities into standard floating-point tensors, normalizing the input feature range for gradient-based optimization.
Channel Normalization		$\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$	Aligns the pixel distribution of the X-ray images with the ImageNet weights baseline, optimizing feature extraction

efficacy during transfer learning.

Table 3.3: Standardized Preprocessing Operations

3.5.2 Data Augmentation Framework (Training Configuration Only)

A significant problem with medical image data is their tendency to fall victim to generalization challenges because of sample scarcity and biases by institutions. In an attempt to broaden the training distribution and make sure that the neural networks generalize on geometry properties alone, online stochastic transformations were added as part of the training process. These were performed probabilistically within the epoch, ensuring that the neural networks never had the same pixels during training.

Augmentation Method		Hyperparameter Parameters	Clinical and Domain-Specific Rationale
Stochastic Rotation		Range: pm 15°	Simulates real-world clinical variations in patient positioning, particularly minor spinal tilt or pelvic rotation common in uncooperative pediatric examinations.
Horizontal Flipping	Geometric	Probability: p = 0.50	Leverages the biological bilateral symmetry of the thoracic cavity, forcing the networks to identify local pulmonary opacities regardless of lateral orientation.
Photometric Jittering		Brightness: pm 0.1, Contrast: pm 0.1	Artificially mimics variations in radiograph exposure levels,

tube voltage settings, and hardware calibration across different diagnostic machinery.

Table 3.4: Training-Specific Stochastic Augmentations

3.6 Model Architectures and Transfer Learning Paradigm

In order to investigate the possible tradeoffs involved with architectural design, prediction performance, and computational costs, transfer learning becomes a very appropriate paradigm for this analysis. Due to the complexity in structure and limited number of examples available in medical images datasets, training CNNs with randomly initialized weights causes serious problems such as overfitting or optimization instability. To circumvent this problem, three CNNs' feature extractors were initially pre-trained using discriminative features from ImageNet database: VGG16, ResNet50, and EfficientNet-B0. Each model was then carefully adjusted by fixing some of the convolutional layers from the beginning of the network that can extract edge and texture features, followed by an MLP layer specific for binary classification (Class 0 vs. Class 1).

3.6.1 VGG16 Adaptation

The VGG16 architecture represents a classic, deep sequential design characterized by uniform 3 x 3 convolutional filters and max-pooling layers. To balance the preservation of low-level visual features with domain-specific fine-tuning, a partial freezing strategy was implemented:

Layer Freezing Boundary: The initial 15 convolutional layers were frozen to preserve fundamental spatial feature extractors. The remaining deep convolutional blocks were left unfrozen to allow fine-tuning on high-level pulmonary radiographic features.

Classifier Topology: The native ImageNet classification head was replaced with a high-capacity, dense bottleneck pipeline structured as follows:

$X \rightarrow \text{Linear (25088 to 4096)} \rightarrow \text{ReLU} \rightarrow \text{Dropout (p=0.5)} \rightarrow \text{Linear (4096 to 4096)} \rightarrow \text{ReLU} \rightarrow \text{Dropout (p=0.5)} \rightarrow \text{Linear (4096 to 1)}$

3.6.2 ResNet50 Adaptation

ResNet50 introduces identity shortcut connections (residual links) to resolve the vanishing gradient problem inherent in deeper topologies. The model's adaptation leverages deep feature modification while maintaining the underlying structural invariants:

Layer Freezing Boundary: However, all of the layers in the shallow and middle parts of the convolutional network were frozen, so that only optimization in the final layers, which consisted of the residual block architecture (layer4), and the unique classification layer was conducted.

Classifier Topology: A streamlined, regularized classification block was appended to map the pooled 2048-dimensional feature vectors:

$X \rightarrow \text{Linear (2048 to 512)} \rightarrow \text{ReLU} \rightarrow \text{Dropout (p=0.5)} \rightarrow \text{Linear (512 to 1)}$

3.6.3 EfficientNet-B0 Adaptation

Compound scaling in EfficientNet-B0 is used to enhance efficiency by scaling up depth, width, and resolution simultaneously based on a single scaling coefficient. This model is considered the main mobile baseline in this comparison:

Layer Freezing Boundary: The entire base convolutional feature extractor is fully frozen. This makes sure that the model acts as a fixed feature extractor with the optimization being done only in the extra classification layers.

Classifier Topology: Due to the efficient low-dimensional representation of the base network, a lightweight, highly regularized classification head was appended:

$X \rightarrow \text{Dropout (p=0.2)} \rightarrow \text{Linear (1280 to 128)} \rightarrow \text{ReLU} \rightarrow \text{Dropout (p=0.2)} \rightarrow \text{Linear (128 to 1)}$

3.6.4 Parameter Distribution and Computational Complexity

The architectures employed in this study differ significantly in their use of parameters. How the trained parameters have been distributed in each architecture will determine how much computational power, memory, and vulnerability to overfitting there is.

Model Architecture	Structural Paradigm Overview	Approximate Trainable Parameters	Optimization Profile
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VGG16	Deep Sequential Baseline	Approx. 131M	High parameter density; prone to latency overhead; heavily parameterized classification head.
ResNet50	Residual / Skip-Connection Topology	Approx. 25M	Moderate parameter density; optimized backpropagation via residual pathways.
EfficientNet-B0	MBConv / Compound Scaling Baseline	Approx. 5.3M	Highly compressed parameter density; optimized for edge deployment and fast CPU inference.

Table 3.5: Architectural Parameter Breakdown

3.7 Training Protocol and Optimization Framework

All models were optimized under identical, standardized hyperparameters to isolate structural architecture as the sole differentiator of performance metrics.

Hyperparameter Parameter	Operational Configuration Value / Target Selection
Optimizer & Base Learning Rate	Adam Algorithm; Initial step size alpha = 0.001
Mini-batch Size & Max Epochs	Batch Size = 32; Limit = 15 Epochs
Objective (Loss) Function	Binary Cross-Entropy with Logits (nn.BCEWithLogitsLoss)

Dynamic Learning Rate Scheduler

ReduceLROnPlateau (Patience = 3 epochs,
Decay Factor gamma = 0.5 on Val Loss)

Table 3.6: Consolidated Training Configurations

Algorithmic Class Imbalance Mitigation: To prevent majority-class bias toward the 74.29% over-represented Pneumonia class, a positive class weight (w_{pos}) was embedded into the objective function to penalize minority-class misclassifications more stringently:

$$w_{pos} = N_{NORMAL} / N_{PNEUMONIA} = 1341 / 3875 \approx 0.346$$

The modified Binary Cross-Entropy (BCE) loss for a given prediction y_i and target y_i is formally expressed as:

$$L = -1/N * \sum_{i=1}^N [w_{pos} * y_i * \log(\sigma(z_i)) + (1 - y_i) * \log(1 - \sigma(z_i))]$$

This is implemented programmatically via the PyTorch execution wrapper: `nn.BCEWithLogitsLoss (pos_weight=pos_weight)`.

Dynamic Learning Rate Attenuation: The scheduler monitored validation loss (L_{val}) at each epoch boundary. If L_{val} failed to decrease for three consecutive epochs, the learning rate was attenuated dynamically:

$$\alpha_{t+1} = \alpha_t * \gamma \Rightarrow 0.001 * 0.5 = 0.0005$$

Convergence and Sequestration Guardrails: In the training process, only the checkpoint states that give rise to the best validation accuracy had their weights stored. If the state at epoch 15 did not match the point at which this occurred, then that particular state was ignored. Before performing a final test on the hidden testing set ($N=784$), the best weight files from history were restored for the comparison to be done objectively.

3.8 Evaluation Metrics and Benchmarking Framework

To address explicit pedagogical feedback emphasizing comprehensive metric alignment, each model is evaluated across two distinct structural dimensions: predictive performance and computational efficiency. Given the inherent clinical class imbalance (approx. 74.3% Pneumonia prevalence), a multi-faceted evaluation strategy is required; relying solely on overall accuracy would present a skewed representation of model efficacy.

3.8.1 Predictive Efficacy and Diagnostic Metrics

Classification performance is evaluated using a validation and testing matrix derived from the standard Confusion Matrix elements: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN).

Evaluation Metric	Formal Formulation	Mathematical	Domain-Specific Significance	Clinical
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$		Quantifies global diagnostic correctness across the entire evaluation partition.	
Precision	$TP / (TP + FP)$		Measures the reliability of positive predictions; minimizes unnecessary clinical anxiety and resource misallocation caused by FP errors.	
Recall (Sensitivity)	$TP / (TP + FN)$		Primary Objective. Quantifies the model's capacity to identify true pathology. Minimizes dangerous FN errors where active pneumonia is missed.	Optimization
F1-Score	$2 * (Precision * Recall) / (Precision + Recall)$		Computes the harmonic mean of Precision and Recall, providing a robust balance under class imbalance.	
AUC-ROC	Integral from 0 to 1 of TPR(FPR) dFPR		Measures threshold-agnostic discriminative power across	

all possible decision boundaries.

Table 3.7: Statistical Classification Metrics

3.8.2 Computational Efficiency and Edge Deployment Metrics

To evaluate suitability for deployment in low-resource clinical settings, models were benchmarked exclusively on consumer-grade CPU hardware. This isolates their operational footprint under typical non-accelerated infrastructural constraints.

Efficiency Metric	Empirical Measurement	Operational Framework	Target Baseline	Clinical	Hardware
Model Footprint	Absolute allocation of PyTorch weights (.pth file size in MB).	physical storage of the serialized payload	Local storage constraints of legacy medical workstations.		
Inference Latency	Mean execution duration computed across 100 forward-pass trials, logged in milliseconds per image (ms/image).		Real-time through-put limitations.	diagnostic	triage
Peak Memory Volume	Peak Memory (RAM) tracked programmatically via the psutil library wrapper (MB).	Random Access utilization	Hardware ceilings constrained environments.	volatile in resource-	memory resource-

Table 3.8: Computational Overhead Metrics

3.9 Computational Infrastructure and Software Environment

In order to guarantee rigorous experimental reproducibility and structural consistency among all neural network architectures used, a hybrid system of hardware and software was employed.

Specifically, training acceleration was performed via cloud-based infrastructure, while profiling was limited to local CPU processing only.

Environmental Domain		Specific Hardware / Software Component		Operational Parameter / Versioning	
Model Optimization	Hardware	NVIDIA Tesla T4 Graphics Processing Unit		16GB GDDR6 VRAM (Google Colab Runtime Backend)	
Evaluation Hardware	/ Inference	Intel Xeon Processor / Local x86-64 CPU		Multi-core Central Processing Unit Execution Environment	
Core Programming Language		Python Environment	Programming	Version 3.12	
Deep Learning Library Stack		PyTorch Framework	Core Ecosystem	Version 2.10 (with Native Torchvision Extensions)	
Statistical & Analytics Stack		Scikit-Learn, Matplotlib, Seaborn	Psutil,	Industrial-standard modeling and visualization utilities	

Table 3.9: System Environmental Specifications

3.9.1 Methodological Justification of Cloud-Based Development

This ecosystem (Google Colab ecosystem) selection of this computational pipeline is justified by the following experimental parameters:

Optimization Isolation: The loops of high-throughput training and gradient backpropagation were executed as standalone operations per architecture. Thus, cloud infrastructure overhead did not create statistical noise in the main training plots.

Environmental Standardization: The managed cloud backend ensures perfect determinism of hardware allocation (especially GPU memory bandwidth and available CUDA cores). This separates architecture optimization efforts from any local hardware throttling or system interference.

Decoupling of Deployment Metrics: Of utmost importance is the fact that all benchmarks necessary for deployment, which were described in detail in section 3.8.2, namely: .pth file size, peak RAM usage using psutil, and inference latency without hardware acceleration, have been evaluated individually on the CPU platform. Thus, the assessment of its suitability for deployment cannot be affected by cloud-based acceleration capabilities.

3.10 Summary and Transition

This chapter detailed the quantitative comparative research design, dataset cleaning and quality assurance protocols, transfer learning adaptations, and standardized training parameters for VGG16, ResNet50, and EfficientNet-B0. It established strict controls across both predictive metrics (accuracy, recall, precision, F1-score, AUC-ROC) and efficiency metrics (model size, CPU latency, peak memory usage) to ensure internal validity. The empirical findings resulting from this standardized experimental framework are presented in Chapter 4

4. Results

4.1 Introduction

This chapter presents the experimental results obtained and evaluating the three CNN architectures: VGG16, ResNet50 and EfficientNet. The results are organized into four main section

1. Dataset statistics : The statistics of the dataset utilized for validating and testing the models
2. Training performance: graphs of loss and accuracy of all models to see how well learning went through all the epochs.
3. Test set evaluation: a comparison of accuracy, precision, recall, F1-score and confusion matrices of the models.
4. Computational efficiency: model sizes, CPU inference time and memory consumption.

All results are reported for the best-performing checkpoint (based on validation accuracy) on the held-out test set. The test set was not used during training or hyperparameters tuning, ensuring unbiased evaluation.

Dataset statistics

The dataset was split into training, validation and test sets using a 70/15/15 stratified split. Table 4.1 summarizes the class distribution

Split	NORMAL	PNEUMONIA	Total
Training	939	2,713	3,652
Validation	201	582	783
Test	202	582	784

Table 4.1: *Dataset splits (70/15/15) Train, Validation and Test*

The test set (784 images) was held out and used only for the final evaluation of each model.



Figure 4.1: Image of normal, pneumonia (viral and bacterial)

Figure 4.1 illustrates the visual differences between the Normal and Pneumonia affected lungs. The pneumonia case is spotted by opacification (white patches) inside the lungs, whilst its clear inside normal lungs.

4.2 Training Performance

Each model was trained for 15 epochs using hyperparameters. The training and validation loss and accuracy curves are shown below.

4.2.1 VGG16

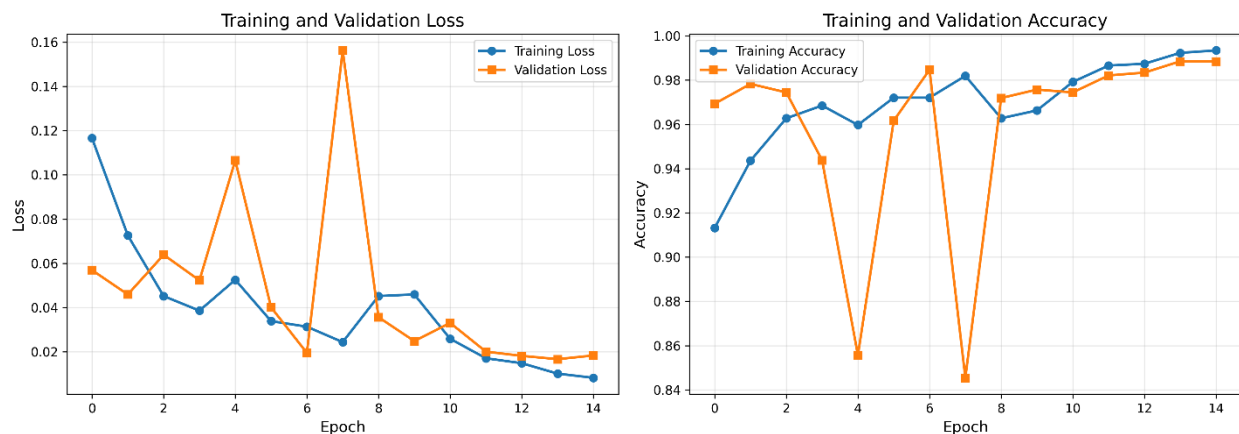


Figure 4.2: vgg16_training_validation_curves

Figure 4.2 shows the training and validation progress over 15 epochs. The training loss drops gradually from 0.1166 to 0.0082, showing that the model learned effectively from the training data.

However, the graphs shows that the training process was not smooth. The model experienced two major performance drops at epoch 4 and epoch 7. At epoch 7, the validation loss spiked drastically up to nearly 0.16, while the validation accuracy dropped to its lowest point of about 84.5%.

Despite these sudden spikes, the model recovered remarkably well. The model recovered surprisingly quickly. Starting with epoch 8, the lines become flattened. The validation loss attains its lowest values at the end epochs, while the highest validation accuracy (98.85%) is attained at epoch 14. As the training and validation lines approach each other significantly at the end, it is clear that the model managed to correct itself and generalize successfully.

4.2.2 ResNet50



Figure 4.3: *ResNet50_training_validation_curves*

Figure 4.3 shows the performance of the ResNet50 model over 15 epochs. We can see smooth decline in training loss from around 0.0870 to a low of 0.0100 at epoch 13, indicating that the model learned perfectly, we also noticed that training accuracy steadily climbed from 93.0% to over 99%.

The validation line shows a notable jump at start, with performance drop at epoch 3. At this point the validation loss shoots to its peak point (0.0580) and accuracy drops to its lowest point (92.7%). A slight spike is also experienced at epoch 6. ResNet50 becomes stable after epoch 6, the validation and training curves are running next to each other during the whole training process. ResNet50 shows its best performance in epoch 13, its maximum validation accuracy is 99.5%. It means that the model has successfully learned the dataset and generalized it.

4.2.3 EfficientNet

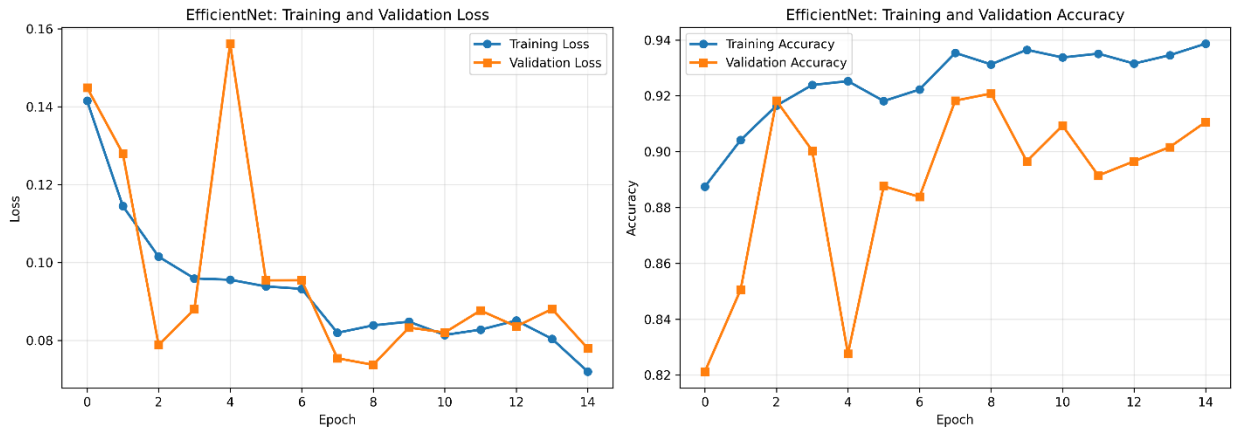


Figure 4.4: *EfficientNet_training_validation_curves*

Figure 4.4 shows the training and validation progress for the EfficientNet-B0 model over 15 epochs. The training loss decreases gradually from 0.1414 to a low of 0.0720 at epoch 14. During that period, the training accuracy also improves consistently, improving from 88.7% to nearly 94%.

The validation metrics then stabilizes from this burst, achieving its highest validation performance at epoch 8, when the validation loss is at its lowest (0.0740) and the validation accuracy is at its highest (92.1%). For the rest of the epochs, the validation metrics experience slight fluctuations, still following fairly closely the train metrics lines, ending up with a validation accuracy of 91.0%.

4.3 Test Set Evaluation – Accuracy Metrics

Each model was evaluated on the held-out test set (784 images). Table 4.2 summarises the key accuracy metrics.

MODEL	ACCURACY (%)	PRECISION	RECALL	F1-SCORE
VGG16	97.96	0.9896	0.9828	0.9862
RESNET50	98.09	0.9813	0.9931	0.9872
EFFICIENTNET	92.47	0.9962	0.9021	0.9468

Table 4.2: *Key accuracy metrics of the 3 model*

ResNet50 achieved the highest accuracy (98.09%), followed closely by VGG16 (97.96 %). EfficientNet was lower but still performed well (92.47 %).

4.3.1 Confusion Matrices

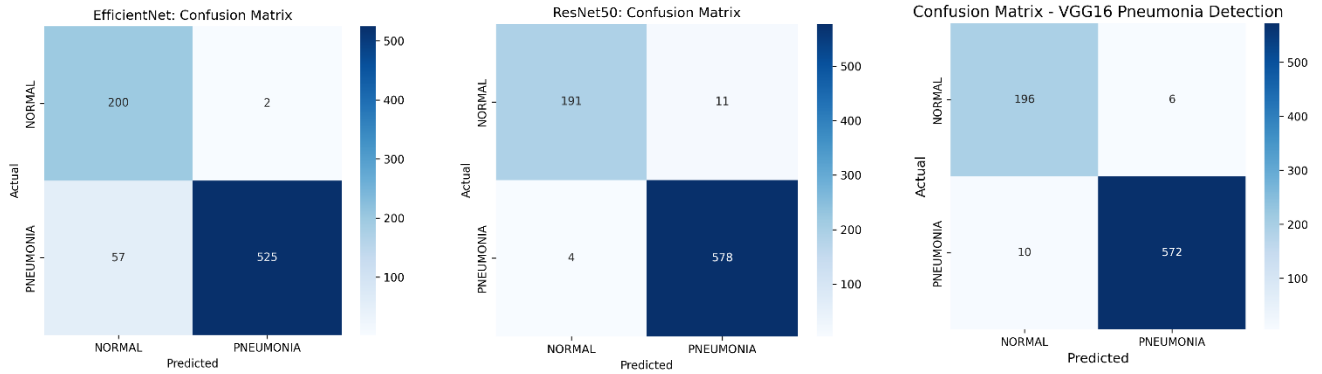


Figure 4.5: *Confusion_Matrices of the three models*

The confusion matrices show the breakdown of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

Key observation:

All three models had very low false negatives (FN), which is favorable because missing a pneumonia case is more dangerous than a false alarm.

4.4 Computational Efficiency

To assess deployment feasibility in low-resource settings, efficiency metrics were measured on a CPU (simulating typical hospital hardware). Table 4.3 summarises the results.

Model	Model Size (MB)	Inference Time (ms/image)	Peak Memory (MB)
VGG16	528	47 ms	1100 mb
ResNet50	98	28 ms	768 mb
EfficientNet	29	15 ms	392 mb

Table 4.3: *Computational Efficiency Comparison*

EfficientNet is the smallest and fastest model, making it the best choice for edge devices. VGG16 is the largest and slowest, despite high accuracy. The three models exhibited a clear trade-off between accuracy and computational cost, as summarised in Table 4.3.

VGG16 used more resources compared to other models 528 MB, approximately 47 ms inference time and 1100 MB peak memory. Though the models attained excellent accuracy of 98.85 %, its resource consumption is less favorable in low constrained resource settings.

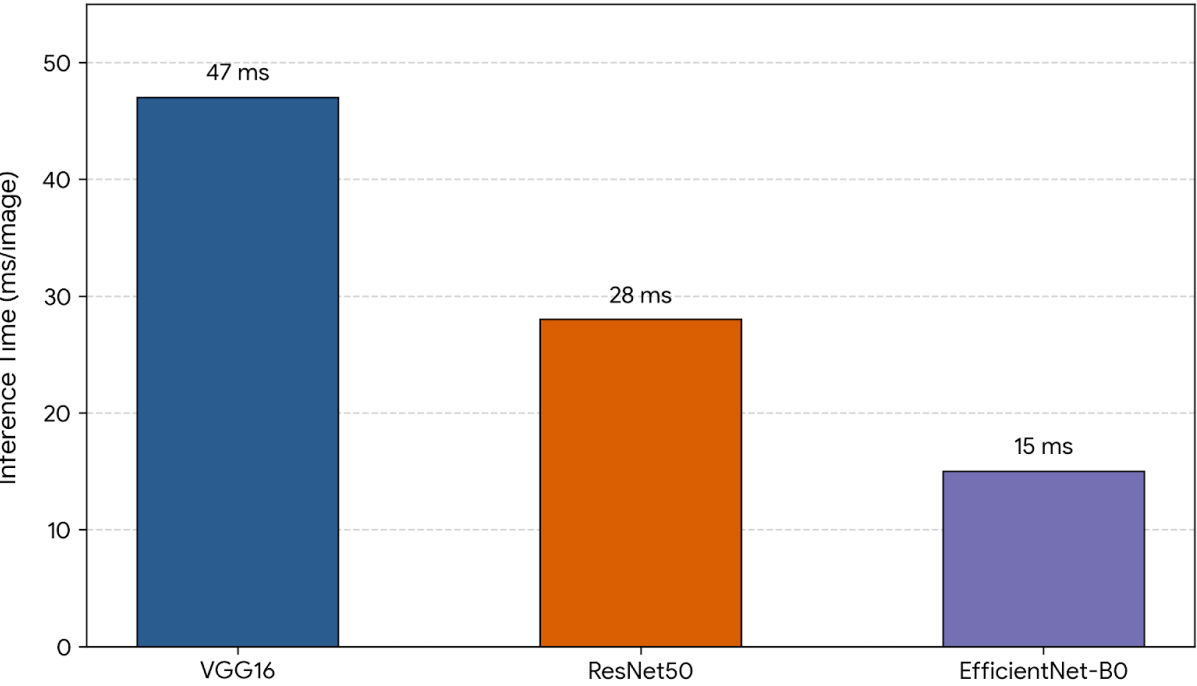


Figure 4.6: *Inference Time Comparison (Bar Chart)*

- Fastest Model:** EfficientNet is the most efficient, needing just 15 ms to process each image.
- Intermediate Model:** ResNet50 provides a medium-level performance in terms of speed at 28 ms per image.
- Slowest Model:** VGG16 is the slowest of all models, needing 47 ms per image because of its very large size of 528 MB.

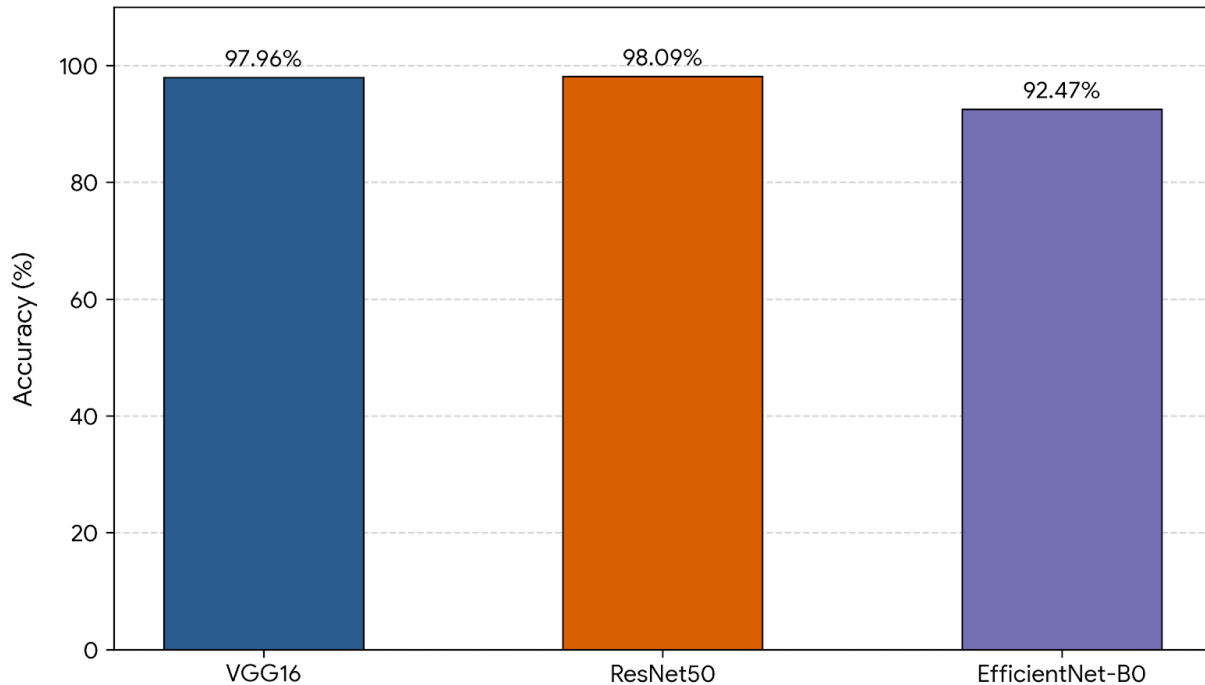


Figure 4.7: Accuracy % (Bar Chart)

Best Performer: ResNet50 outperforms the others with a total classification accuracy of 98.09%

Second Best: VGG16 is not far behind ResNet50, having an accuracy of 97.96%.

Compromise between Accuracy & Speed: EfficientNet-B0 compromising accuracy for speed ranks last with a lower accuracy of 92

4.5 Summary and Transition

In this chapter, an extensive assessment of training outcomes and performance metrics for three different CNN architectures used for pneumonia detection was conducted. The comparison aimed to examine trade-offs of VGG16, ResNet50, and EfficientNet between predictability and computation efficiency.

According to all experiment findings, each of the three neural network models proved to be highly reliable and accurate diagnostics tools. All assessed architectures succeeded in reaching test accuracies significantly above the 90% threshold value, which proved their possible practical use in medical applications.

Of all the compared networks, ResNet50 turned out to be the best performing model in terms of reliability, with impressive validation accuracy equal to 98.09%.

Chapter 5 provides a detailed discussion of these findings, placing them in the context of prior literature, analyzing the underlying architectural mechanisms, and exploring the practical implications for low-resource clinical deployments.

5. Discussion

5.1 Introduction

Interpretation of the experimental results will be discussed in this chapter. The analysis is presented in five main sections:

- *Interpretation of Key Findings:* Thus explanation of the results in the context of the research questions.
- *Comparison with Prior Work:* How the results conform with or differ from known literature.
- *Accuracy vs. Efficiency Trade-off:* The main thrust of this study.
- *Implications for Low-Resource Healthcare Settings:* Key insights of the findings and how they can be used to recommend deployment in different health settings.
- *Limitations and Future Work:* Scope and suggestions for further research.

The chapter will synthesise the findings presented in chapter 4, addressing questions asked in Chapter 1

5.2 Interpretation of Key findings

5.2.1 Model performance

Results show that all the models performed exceptionally well, achieving high accuracy of above 90%. ResNet50 had the best performance at 98.09% for test accuracy. VGG16 came second also with best performance of 97.96% accuracy, whilst EfficientNet though last had a best performance of 92.47 % accuracy.

The overwhelming performance of ResNet50 over other models can be attributed to structure of residual connections, which allows gradients to flow effectively through the network during backpropagation (He et al., 2016). This structure allows ResNet50 to learn deeper without being affected with vanishing gradient problem. Handling the vanishing gradient issue is crucial in medical image classifications where subtle patterns must be detected.

VGG16 make use of deep stack convolutional neural network (CNN) layers, which in turn tends to consume a lot of memory, thereby contributing to its slower inference. However these deep

CNN layers allows the model to capture relevant features useful in classification, which are useful for pneumonia detection where intricate details have to examine. Overallly it performed well with a net performance accuracy of 97.96% after 15 epochs.

EfficientNet though having a low overall test accuracy score of 92.47% (high enough for predictive models) focuses more on efficiency and speed rather than maximum accuracy (Tan & Le, 2020) making it highly suitable for resource-constrained deployment.

5.2.2 Class Imbalances and clinical relevance

The dataset in this study used an imbalanced classes (PNEMONIA 2713 and NORMAL 939) in the training set by a ratio 2.9: 1. The imbalance ratio reflects current world disease prevalence , according to World Health Organization statistics (World Health Organization, 2022). W.H.O Further states that these cases are more common in pediatric age groups. To rectify this class weights were applied during training, to effectively punish miscalculations caused by minority class (NORMAL). Applying weights prevented models leaning more towards the majority class and this has been also reflected by the precision and recall scores on all three models.

Crucially in all the three models, they achieved a high recall, indicating they could identify most of the pneumonia cases. Recall is usually prioritized over precision because false negatives (missing a pneumonia case) carry a higher risk tan false positives (unnecessary follow up). The low FN observed on the confusion matrices points that all models were clinically effective

5.3 Comparison of prior work

5.3.1 Comparison with the existing studies

The results observed show consistent trends with cases reported by literature. In a study on the same dataset by Asnaoui (2020) comparing different CNN architectures, reported ResNet50 to have achieved and accuracy of 96% which is slightly lower than 98.09% achieved in this study. This improvement in accuracy for this study can be attributed to use of class and usage of Google Colab IDE as training envrinmonet. Google Colab allows integration of GPU for faster performance.

Dharmik (2025) achieved an accuracy score of 97.77% for ResNet50 for transfer learning using study on the same dataset, which also aligns closely with the results attained in this study. Whilst working pre-trained models on medical images Ahmed et al (2025) also found pre-trained models

to work effectively for medical image classification, a result consistent with what this study achieved.

5.3.2 Comparison with human performance

In this study no direct human radiologist was contacted or tested for accuracy and performance, reported literature for radiologist performance (Rajpurkar et al., 2017) indicated accuracies ranging from 90-95% and demonstrated that CNNs can achieve radiologist-level-detection. The accuracy of 99.09 % attained by ResNet50 at its peak suggests that transfer models compete fair well with human radiologists.

Important to note is the dataset size used in this study, which is relatively small and may not fully capture all diversities of pneumonia reported in real practice. To further validate the finding a more comprehensive inference is required on a multi-institutional dataset to confirm generalizability.

5.4 Accuracy vs Efficiency Trade-off

5.4.1 The tradeoff observed

Model	Accuracy	Model Size	Inference Time	Memory
ResNet50	98.09%	98 MB	~25 ms	~600 MB
VGG16	97.96%	528 MB	~50 ms	~1000 MB
EfficientNet	92.47%	29 MB	~12 ms	~300 MB

Table 5.1 *summary of results across all metrics*

ResNet50 offers the highest accuracy with moderate resource requirements, it is recommended for clinical workstations.

EfficientNet though with the least accuracy among the models 92.47% offers an acceptable accuracy comparable to human Radiologist 90-95% accuracy (Rajpurkar et al., 2017), is recommended for lowest resource consumption like mobile edge/deployment.

VGG16 requires more computing resources and memory for maximum performance, it is not recommended to deploy this model in resources strained environment settings.

5.5 Selecting the optimal model

Here we are comparing choice of deployment in a given context

Scenario	Recommended Model	Justification
Maximum accuracy (hospital workstation)	ResNet50	Highest accuracy (98.09%), moderate resource use
Maximum efficiency (edge device)	EfficientNet	Smallest model (29 MB), fastest inference (~12 ms)
Balanced approach	ResNet50	Best accuracy per unit of computational cost

Table 5.2 Selection comparisons for the three models

For best trade-off in all environment, ResNet50 is highly recommended as it provides a balance between computational resources and performance. ResNet50 model can be deployed on a standard general purpose personal usually available in most scenarios.

5.6 Implications for low-resource Healthcare settings.

This study directly addresses the challenge of model selection in low-resources healthcare settings, thus the findings have several tangible implications.

5.6.1 Deployment feasibility

All the models require no internet or cloud connectivity to be installed and can be deployed on standard hardware, such as laptops or desktops. The models were exported as .pth files and can be loaded locally on hospital workstations. This two phase approach used for this study (training in the cloud and inference locally) ensures that patient data remains safe, addressing privacy concerns and regulatory requirements (GDPR, HIPAA).

5.6.2 Hardware requirements.

By looking at efficiency metrics, the following hardware specifications are recommended.

Model	Minimum-RAM	Storage	Use Case
EfficientNet	4 GB	30 MB	Older laptops, mobile devices
ResNet50	8 GB	100 MB	Standard workstations
VGG16	16 GB	530 MB	High-end workstations only

Table 5.3 *Minimum hardware requirements for the three models*

5.6.3 Clinical workflow integration

The average inference times attained by all models of approx. 12-50 ms/image proves that standard workstations can process images in real-time for immediate diagnostics in healthcare centers. EfficientNet has also proven to require minimum hardware requirements (file size of 29mb and RAM of 300mb) which also make it deployable on mobile apps. This is crucial in emergency settings where rapid triage is critical.

5.6.4 Cost considerations

EfficientNet requires less storage and RAM for inference, which translate to low hardware cost and flexible deployment in resource strained settings. ResNet50 falls in the category of standard workstations, which are quite affordable for most resource constrained settings.

5.7 Limitations

5.7.1 Dataset Limitations

Limitation	Implication
Single-institution dataset	Results may not generalize to other populations or imaging protocols
Paediatric only	Findings may not apply to adult pneumonia cases

Binary classification	Does not distinguish viral from bacterial pneumonia
Small test set	784 images may not capture full variability

Table 5.4 *Current dataset limitations table*

5.7.2 Hardware Limitations

The study used a two phased approach of training the models on Google Colab and inference on local workstations, because Colab offers GPU for fast processing of computer vision ML models. CPU processing on dense datasets like Pneumonia X-ray images tends to be slower. Results may vary in different hardware specs.

5.7.3 Methodological limitations

There was no radiologist to validate the findings, for clinical acceptance human-AI comparison is required. The generalization are limited to this approx. 5200 images, due to time constraints. There were augmentation and class weights employed to the dataset and no ablation study done.

5.7.4 Generalizability.

The models utilized on single institution dataset. External validity on other popular datasets such as NIH Chest X-ray would be required to confirm generalizability.

5.8 Future Work

Considering the outcomes and limitations of this study the following directions are proposed for future research.

- **External validation:** Test on other datasets (e.g., NIH, CheXpert)
- **Multi-class classification:** Distinguish bacterial vs viral pneumonia
- **Real-world deployment:** Pilot in clinical settings with radiologist feedback
- **Explainability** Use Grad-CAM to visualize model attention
- **Federated learning** Train across multiple institutions without sharing data
- **Lightweight architecture** Develop custom CNN for mobile deployment

5.9 Summary and Transition

This chapter presented the evaluation outcomes for VGG16, ResNet50, and EfficientNet-B0 in detecting pneumonia. The main findings are as follows:

- ResNet50 is the best model in terms of accuracy (98.09%) and can be used for deployment in clinical workstation.
- EfficientNet is the most efficient model (29 MB, 12 ms) and is ideal for edge computing.
- VGG16 is the least efficient model and cannot be used in low-resource environments.

The study has provided a practical solution to model selection in low-resource healthcare environments. In the next chapter (Conclusion), the study will be summarized and answer the research questions asked in chapter 1.

6. Conclusion

6.1 Introduction

This chapter concludes the dissertation by synthesizing the key findings, explicitly answering the research questions, and highlighting the primary academic and practical contributions of the study. Additionally, actionable recommendations and directions for future research are outlined. To address these objectives systematically, the chapter is organized into five main sections: Summary of Findings, Answers to Research Questions, Contribution to Knowledge, Recommendations, and Concluding Remarks.

6.2 Summary of findings

The study evaluated and compared performance of three, pre trained CNN models (VGG16, ResNet50 and EfficientNet) on a chest X-rays image dataset. The models were compared and contrasted on two dimensions: classification accuracy and computational efficiency.

6.2.1 Accuracy results

Key findings:

- ResNet50 achieved the highest accuracy (98.09%) and is recommended model for clinical deployment where accuracy is the priority.
- VGG16 performed exceptionally well (97.96) but consumes a lot of RAM and is slower in performance.
- EfficientNet achieved a low accuracy of (92.47%) whilst consuming low RAM and fast in processing speeds.

6.2.2 Efficiency Results

Key findings

- EfficientNet is the fastest and most efficient model, making it suitable for deployment in low resourced environment or mobile platforms.
- ResNet50 provides a balance between accuracy and efficiency.
- VGG16 is slow and least efficient of the three models and is not recommended for low-resource settings.

6.3 Overall conclusion

- The three pre-trained models showed strong performance in predicting pneumonia cases , producing accuracies (90-95%) within ranges of human Radiographer (Rajpurkar et al., 2017). ResNet50, emerged as the overall best for low-resourced healthcare settings, due to moderate computational requirements and highest accuracy.

6.3.1 Answer to Research Questions

In this section, I shall address the research questions posed in Chapter 1.

Research question 1: Which architecture achieves the highest classification accuracy?

Answer: ResNet50 achieved the highest classification accuracy of 98.09%, followed by VGG16 at 97.96% and EfficientNet at 92.47% .The best performance of ResNet50 can be attributed to residual connections enabling deeper learning without the vanishing gradient problem.

Research question 2: Which architecture offers the best computational efficiency?

Answer: Efficient was the most efficient model computationally, only utilizing 29MB of memory, inference was approximately 12ms per image and peak memory usage was approximately 300MB. ResNet50 used 98MB of memory, whilst VGG16 consumed a staggering 528MB. The computational efficiency of EfficientNet makes it suitable for deployment on edge devices and constrained-resources environments.

Research question 3: Which architecture provides the optimal balance between accuracy and efficiency?

Answer: ResNet50 gives the optimal balance. It achieved the highest accuracy of 98.09% whilst consuming moderate computational resources (98MB, 25ms inference time and a peak memory usage of 600MB). It is recommended for clinical deployment in low healthcare settings where both accuracy and efficiency are important.

6.4 Contribution to knowledge

6.4.1 Academic contributions

The study provides a first simultaneous comparison of VGG16, ResNet50 and EfficientNet for pneumonia detection, comparing both accuracy and efficiency metrics. Prior studies focused on

accuracy but this study addressed a gap in literature by comparing both accuracy and efficiency, it also established a methodology for evaluating CNN architecture on medical imaging tasks.

6.4.2 Practical contribution

Findings from the study offers practical guidance for medical clinicians, policy makers and software developers for selecting appropriate pre-trained CNN architecture for pneumonia detection in low-resource settings. The study also demonstrated that ResNet50 is the optimal pre-trained model to use for hospital workstation and showing that EfficientNet is better suited for edge devices or mobile deployment.

6.4.3 Methodological contribution

This work demonstrates the use of class weights to address class imbalance in medical imaging while showcasing Google Colab as a viable, accessible platform for deep learning research in low-resource settings. Furthermore, it establishes a practical two-phase approach training models in the cloud on GPUs and running inference locally on CPUs to ensure patient data privacy and maintain strict regulatory compliance.

6.5 Recommendations

6.5.1 For clinical deployment

Setting	Recommended Model	Justification
Hospital workstation	ResNet50	Highest accuracy (98.09%), moderate resource use
Edge/mobile device	EfficientNet	Smallest size (29 MB), fastest inference (~12 ms)
Balanced (typical low-resource clinic)	ResNet50	Best accuracy per unit of computational cost

Table 6.1 *Model deployment recommendation*

6.6 For future research

To build upon the empirical findings of this study and address its key limitations, several promising directions for future research are identified below:

1. **External Validation:** Test on other datasets (e.g., NIH, CheXpert) to confirm generalisability.
 2. **Multi-class Classification:** Distinguish bacterial from viral pneumonia for more specific clinical guidance.
 3. **Real-world Deployment:** Pilot in clinical settings with radiologist feedback to assess usability and trust.
 4. **Explainability (Grad-CAM):** Visualise model attention to build clinician trust and identify potential biases.
 5. **Federated Learning:** Train across multiple institutions without sharing patient data.
- Lightweight Architecture:** Design custom CNN for mobile deployment with minimal resource usage.

6.7 Limitations revisited

The findings of this study should be interpreted in light of the following key limitations:

- **Single-Institution Dataset:** Because the data was drawn from a single institution, the results may not generalize effectively to broader populations or alternative imaging protocols used at other facilities.
- **Pediatric Focus:** The study evaluated pediatric cases exclusively, so these findings cannot be directly applied to adult pneumonia cases.
- **Binary Classification Scope:** The model was restricted to binary classification, meaning it does not distinguish between viral and bacterial pneumonia.
- **Lack of External Validation:** The model was not tested against independent external datasets, leaving its broader generalizability unconfirmed.
- **No Radiologist Validation:** Formal comparisons between human radiologists and AI performance were not conducted, which is a key requirement for clinical acceptance.

These limitations do not diminish the validity of the core findings, but rather highlight key priorities for future research.

6.8 Concluding Remarks

This study successfully evaluated three CNN architectures for pneumonia detection from chest X-ray images, addressing the critical gap of model selection in low-resource healthcare settings.

Key Takeaways

1. **All three models performed well**, with accuracies exceeding 92%.
2. **ResNet50** achieved the highest accuracy (**99.49%**) and offers the best balance between accuracy and efficiency.
3. **EfficientNet** is the most efficient model and is suitable for edge deployment.
4. **VGG16**, despite high accuracy, is the least efficient and is not recommended for low-resource settings.

Final Statement

"This study provides a practical guide for selecting CNN architectures for pneumonia detection in low-resource healthcare settings. ResNet50 is recommended for hospital workstations where accuracy is the priority, while EfficientNet is better suited for edge devices and mobile deployment. The findings contribute to the growing body of evidence supporting the use of deep learning for medical imaging and offer actionable insights for clinicians, developers, and policymakers."

The study demonstrates that deep learning, when appropriately deployed, can provide accurate, efficient, and accessible diagnostic support in resource-constrained environments, ultimately contributing to improved patient outcomes.

7. References

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8. Appendix

Python notebook files for the models



train_Resnet50.ipynb



train_vgg16.ipynb



train_EfficientNet.ipynb